

New Hope Lutheran Church

AV Tech System

OBS Studio Setup, Operation, and Rebuild Guide

Companion documentation module for Stream Agent III sd

Prepared: May 19, 2026

Status: Working baseline documentation - update with actual OBS screenshots when available

Sensitive information note

This document intentionally does not include the YouTube stream key, OBS WebSocket password, or camera passwords. Record those securely outside any public GitHub repository or printed public manual.

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1. Purpose and scope

This document explains the OBS Studio part of the New Hope Lutheran Church AV tech system. It is written so that a new volunteer, or a future maintainer, can understand what OBS does, why it is needed, how Stream Agent III sd controls it, and how to rebuild the OBS configuration if the BeelinkOBS computer has to be replaced.

This is not the final YouTube manual. YouTube scheduling, thumbnails, auto-start, auto-stop, and placeholder management should be documented in a separate YouTube-specific guide. This document covers only the OBS side of that relationship.

- OBS = Open Broadcaster Software. In this document, OBS means OBS Studio.
- NDI = Network Device Interface, used here mainly for Proclaim slide/overlay video reaching OBS over the local network.
- RTSP = Real-Time Streaming Protocol, used by the Axis cameras to send camera video/audio streams to OBS.
- RTMP / RTMPS = Real-Time Messaging Protocol / secure RTMP, used by OBS to send the final live program to YouTube.
- ASIO = Audio Stream Input/Output, an optional fallback/legacy audio-driver method. The standard New Hope audio path is embedded Axis camera audio from the mixer feed.
- WebSocket = a network control interface used by Stream Agent III sd to control OBS locally.

2. What OBS Studio is

OBS Studio is the live production and encoder application. It builds the final program picture from scenes and sources, mixes or carries audio sources, encodes the result, and sends it to YouTube Live. OBS is the point where camera video, slide overlays, intro media, and stream audio become one outgoing live program.

OBS scenes and sources are the core of the configuration. A scene is a complete layout; a source is one item inside that layout, such as a camera stream, media file, NDI overlay, browser source, or audio source.

Important mental model

Stream Agent III sd does not stream video to YouTube by itself. It tells OBS what to do. OBS is still the encoder and the program-output engine.

3. Why New Hope uses OBS

New Hope needs a flexible live-production program because the service stream is not a single fixed camera. The system must combine two Axis PTZ cameras, Proclaim slide overlays, an introduction clip, audio, and YouTube output. OBS provides the scene and source structure needed to assemble and encode this final program.

- OBS can receive multiple video sources and layer them in the correct order.
- OBS can stream to YouTube using a persistent church stream key.
- OBS exposes a WebSocket control interface so Stream Agent III sd can automate start/stop, scene switching, media restart, screenshots, and audio controls.
- OBS allows the same scene collection to be backed up and later restored if the PC fails.
- OBS is familiar to many streaming volunteers and has a large support base.

4. How OBS fits into the AV tech system

OBS runs on the BeelinkOBS streaming PC. It sits between the church AV network and YouTube. Cameras, Proclaim, and audio arrive at or are controlled through OBS; OBS then sends the final stream to YouTube.

OBS Studio in the New Hope AV Tech System

OBS is the central encoder and production canvas. Stream Agent III sd controls OBS; OBS sends the final program to YouTube.

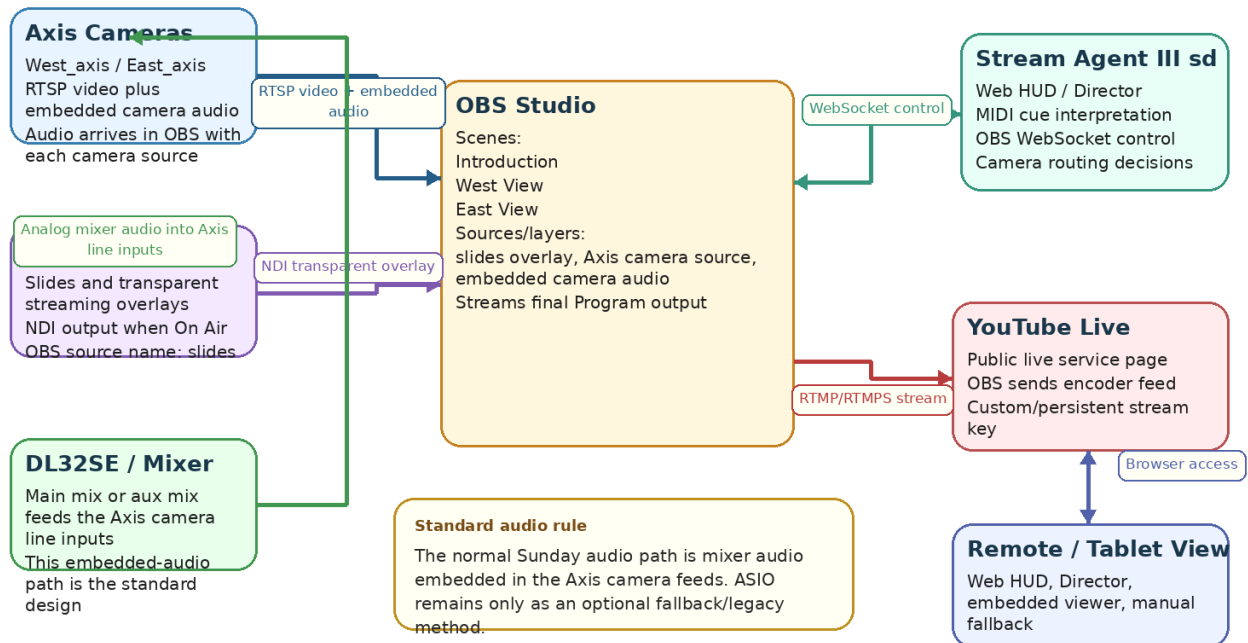


Figure 1 - OBS as the central hub in the New Hope AV tech streaming system.

5. How Stream Agent III sd controls OBS

Stream Agent III sd connects to OBS through the OBS WebSocket interface, normally on the same PC using localhost.

Item	Current/expected value	Purpose
OBS host	127.0.0.1	Stream Agent connects locally on the BeelinkOBS PC.

OBS WebSocket port	4455	Default OBS WebSocket port used by current app settings.
OBS password	Redacted / configure to match OBS	Required only if OBS WebSocket authentication is enabled.
Expected profile	NHLC<space>	Used by the app profile safety check when enabled.
Web HUD port	8765	Browser interface served by Stream Agent, not by OBS.

Stream Agent uses OBS control for these actions:

- Start stream and stop stream.
- Toggle OBS recording.
- Switch Program scene to Introduction, West View, or East View.
- Set a scene transition, including Cut for intro or app-directed operations when configured.
- Restart the Intro_Video media source and watch for the media-ended state.
- Read OBS screenshot images for the Director preview page.
- Adjust OBS input volume for the configured audio target(s).
- Use the ASIO sync-offset controls only if the optional ASIO fallback mode is deliberately selected.

Why exact names matter

Stream Agent addresses OBS items by name. If an OBS scene or source is renamed, Stream Agent must be updated or the app will fail to find that scene/source.

6. Current OBS naming requirements

The following names are the key rebuild values. They should be treated as part of the interface contract between OBS and Stream Agent III sd.

OBS item	Required / expected name	Used by
Scene	Introduction	Intro playback sequence
Scene	West View	West camera program scene
Scene	East View	East camera program scene
Media source/input	Intro_Video	Intro media restart and state monitoring
Camera source/input	West_axis	West camera video, Director preview, Axis embedded audio target
Camera source/input	East_axis	East camera video, Director preview, Axis embedded audio target
Proclaim overlay source	slides	Proclaim lower-third/overlay visual layer in live scenes

Optional ASIO fallback source	ASIO_audio	Only used when AUDIO_MODE = asio_shared
Expected OBS profile	NHLC<space>	Profile safety check and correct stream destination; exact name includes the trailing space; exact name includes the trailing space

7. Software requirements and plugins

The following software/plugins are part of the OBS side of the system. Versions should be recorded when the system is stable. Do not update OBS or plugins immediately before a service unless there is a verified rollback plan.

Software / plugin	Required for	Notes
OBS Studio 32.x or current stable working version	Main encoder and scene/source host	OBS should be started before Stream Agent or by Task Scheduler before service.
OBS WebSocket	Stream Agent control	Built into OBS Studio 28 and newer; normally no separate install is needed.
DistroAV / NDI plugin	Proclaim overlay source in OBS	Formerly known as OBS-NDI. Needed if Proclaim overlays are captured into OBS by NDI.
NDI runtime/tools	NDI transport support	Install alongside DistroAV if required by the plugin/version.
OBS ASIO plugin	Optional ASIO fallback only	Needed only if the optional ASIO fallback mode is deliberately used with an ASIO source named ASIO_audio.
Proclaim	Slide/overlay source	Proclaim must be On Air and sending the NDI source expected by OBS.
Stream Agent III sd	Automation and Web HUD	Controls OBS through WebSocket; does not replace OBS.

8. OBS scenes and source design

The scene collection should be simple and predictable. The current Axis-only Stream Agent III sd design expects three main program scenes.

Required OBS Scene and Source Layout

Exact names matter: Stream Agent III sd addresses scenes and sources by name. Embedded Axis audio is the standard audio path.

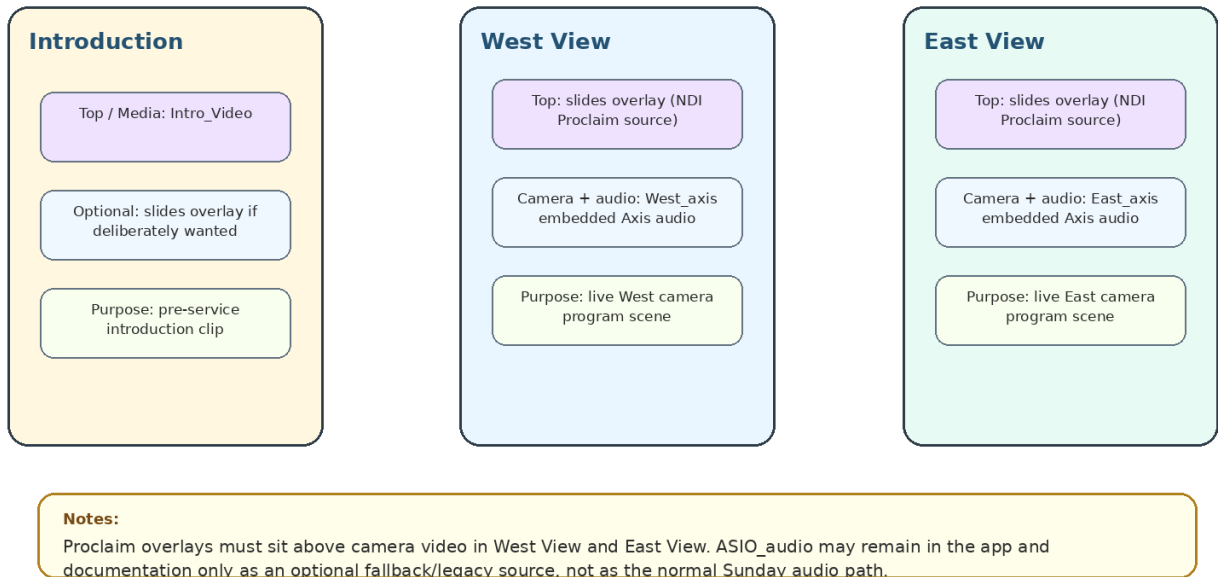


Figure 2 - Required scene/source layout and layer order.

8.1 Introduction scene

Layer / source	Name	Purpose / notes
Media source	Intro_Video	The introduction clip. Stream Agent restarts this source after stream start and waits for it to end.
Audio source	Normally none or embedded camera-audio path; ASIO_audio only if fallback ASIO is selected	If the intro video has its own audio, test carefully. Standard service audio is embedded in the Axis camera sources.
Optional overlay	slides	Only if a Proclaim overlay is deliberately wanted over the intro. Usually not required.

8.2 West View scene

Layer order	Source name	Purpose / notes
Top overlay	slides	Proclaim NDI overlay. Must be above the camera layer.
Camera video	West_axis	Axis RTSP source for the West camera.
Audio target	West_axis embedded audio standard; ASIO fallback	Embedded audio is standard; ASIO fallback only.

8.3 East View scene

Layer order	Source name	Purpose / notes
Top overlay	slides	Proclaim NDI overlay. Must be above the camera layer.
Camera video	East_axis	Axis RTSP source for the East camera.
Audio target	East_axis embedded audio standard; ASIO fallback	Embedded audio is standard; ASIO fallback only.

9. Camera source setup

The Axis cameras are normally added to OBS as network media sources. The exact RTSP URLs should be recorded securely in the local AV tech setup notes. The app uses the source names West_axis and East_axis to request screenshots and control audio levels in embedded-audio mode.

Camera	IP address	OBS source name	Scene
West Axis M5525-E	192.168.88.2	West_axis	West View
East Axis M5525-E	192.168.88.3	East_axis	East View

- Use the exact OBS source names shown in this table unless Stream Agent config is also updated.
- Prefer stable wired Ethernet through the ER605/PoE switch AV LAN.
- If RTSP transport options are available, TCP is usually preferred for reliability over the local wired network.
- Do not add extra direct camera preview streams unless needed; too many camera clients can increase load and latency.

10. Proclaim overlay / NDI setup

Proclaim provides the church slides and overlay content. In the current working design, OBS receives the Proclaim overlay through an NDI source and places it above the camera source in each live scene.

OBS source	NDI source seen when Proclaim is On Air	Used in scenes
slides	BEELINKOBS (Proclaim - proclaim overlays)	West View and East View

- The OBS source name should remain slides.
- If the overlay disappears, first confirm Proclaim is open and On Air.
- Confirm DistroAV/NDI is installed and that the Proclaim NDI source is visible to OBS.
- Layer order matters: slides should be above West_axis or East_axis in the scene source stack.

11. Audio design in OBS

Standard audio decision: embedded audio from the mixer into the Axis cameras is now the favored and assumed normal Sunday sound mechanism. ASIO support remains in the app and documentation only as an optional fallback/legacy path.

The standard New Hope audio model is Axis embedded audio: the mixer feed enters the Axis cameras and arrives in OBS as audio on West_axis and East_axis. The app still contains ASIO support as an optional fallback/legacy mode, but that is not the normal Sunday configuration.

Mode	Stream Agent value	OBS requirement	When to use
Axis embedded audio - standard	AUDIO_MODE = axis_embedded	Audio is carried by West_axis and East_axis media sources. The Web HUD audio fader controls those OBS inputs.	Current favored and assumed standard. The Sync page is hidden because audio and video travel together in the camera source.
Shared ASIO audio - optional fallback	AUDIO_MODE = asio_shared (fallback only)	One OBS audio source named ASIO_audio is added only if the fallback ASIO mode is deliberately used.	Optional fallback/legacy only. Use only if the embedded Axis audio path is deliberately abandoned and a separate audio interface/mixer path feeds OBS directly.

11.1 Optional ASIO_audio fallback add-existing rule

If the optional ASIO fallback mode is deliberately used, create one OBS source named ASIO_audio and add it to West View, East View, and usually Introduction using Add Existing. Do not create separate independent ASIO sources with similar names unless the app and documentation are updated. For the standard embedded Axis audio design, this ASIO source is not required.

Why this matters

One shared ASIO_audio source keeps the fallback ASIO mode consistent if it is ever needed. The standard embedded Axis audio design does not depend on ASIO_audio.

12. OBS profile, scene collection, and output settings

Exact profile warning: the current OBS profile name is NHLC followed by a trailing space. In this document it is written as NHLC<space> so the invisible trailing space is not missed during a rebuild.

OBS separates output/stream settings from scene/source layouts. A Profile stores many output-related settings. A Scene Collection stores scenes and sources. Both must be backed up.

Item	Current / expected	Notes
OBS Profile	NHLC<space>	Used by Stream Agent profile safety. The trailing space is part of the exact OBS profile name.
Scene Collection	Verify current church scene collection name	Should contain Introduction, West View, East View. Record final name after confirming.
Output sample rate	48 kHz expected	Should match the church audio path where practical.
Streaming service	YouTube Live via RTMP/RTMPS	Use the persistent New Hope stream key; never publish the key.
Transition	Fade 400 ms currently reported acceptable; Cut may be forced by app for intro/certain operations	Avoid changing shortly before service without test.

The final confirmed OBS settings should be captured later with screenshots from Settings -> Stream, Output, Audio, Video, Advanced, and WebSocket Server Settings.

13. YouTube Live interworking - OBS side

OBS sends the final program feed to YouTube using the configured stream server and stream key. New Hope uses a recurring/persistent stream-key workflow so the OBS key does not normally change each service. YouTube should have its own detailed document; this section records only the OBS-facing facts.

- The YouTube stream key is sensitive. Treat it like a password.
- The current New Hope approach is to use a persistent/custom stream key in OBS and create/schedule YouTube live placeholders separately.
- YouTube auto-start/auto-stop settings determine whether a scheduled stream starts/stops when OBS begins/stops sending video.
- Stream Agent starts and stops OBS; YouTube must already be prepared to accept the stream.

- The Stream Agent Web HUD includes a View Live button pointing to the New Hope YouTube live page.

New Hope public live URL used by Stream Agent:

<https://www.youtube.com/@NewHopeLutheranChurchRegina/live>

14. Normal Sunday operating sequence

1. Confirm the ER605 AV LAN, PoE switch, Axis cameras, and BeelinkOBS are powered.
2. Start or confirm OBS is running. Task Scheduler normally handles this at login.
3. Confirm OBS is on the correct Profile and Scene Collection.
4. Confirm the Program scene is safe, typically West View or Introduction before service start.
5. Start Proclaim and confirm it is On Air when overlays are needed.
6. Start or confirm loopMIDI and Stream Agent III sd are running.
7. Open the Stream Agent Web HUD and Director page from the tablet or browser.
8. Confirm Director previews show West and East cameras, or troubleshoot before going live.
9. Confirm audio meters respond and the Web HUD fader is at the intended level.
10. Use timer, Web HUD, or MIDI note 60 to start stream. Stream Agent should start OBS and play Introduction if enabled.
11. During the service, let Proclaim MIDI cues drive normal camera changes. Use Director for manual fallback.
12. At the end, use Web HUD or MIDI note 61 to start the stop countdown and stop OBS streaming.

15. Rebuilding OBS after a PC failure

This is the core reconstruction procedure. It assumes the replacement PC is already on the AV LAN and can reach the Axis cameras and internet.

15.1 Best case: restore from backup

13. Install OBS Studio and the required NDI components. Install ASIO components only if the optional fallback ASIO audio path is deliberately being used.
14. Copy the backed-up OBS profile and scene collection into the new user profile, or import them through the OBS menus.
15. Copy the Stream Agent III sd folder and config_overrides.json.
16. Install Proclaim, Master Fader, loopMIDI, and other required AV software.
17. Set up Task Scheduler auto-launch tasks.
18. Start OBS and verify the profile, scene collection, source names, and WebSocket settings.
19. Run a private YouTube test before relying on the rebuilt system for Sunday service.

15.2 Manual rebuild if no OBS backup exists

20. Install OBS Studio.
21. Install DistroAV/NDI support if Proclaim overlay is used through NDI.
22. Install OBS ASIO plugin only if the optional ASIO fallback mode will be used.

23. Create or select the OBS Profile named NHLC<space>. The actual OBS name includes the trailing space. The actual OBS name includes the trailing space.
24. Configure Settings -> Stream for YouTube using the New Hope persistent stream key. Do not document the key in public files.
25. Create a scene collection for the church service setup.
26. Create scenes named Introduction, West View, and East View exactly.
27. Add media source Intro_Video to Introduction.
28. Add Axis RTSP source West_axis to West View.
29. Add Axis RTSP source East_axis to East View.
30. Add the Proclaim NDI overlay source as slides to West View and East View above the camera layers.
31. If using axis_embedded audio, confirm West_axis and East_axis audio meters appear and the Web HUD fader controls them.
32. Only if using the optional asio_shared fallback mode, create one ASIO_audio source and add it as existing to all relevant scenes.
33. Enable/configure OBS WebSocket on port 4455 and match the password in Stream Agent if authentication is used.
34. Open Stream Agent Web HUD and verify OBS status is online.
35. Use Director to verify previews and issue a safe scene cut test.
36. Run a private stream/recording test to confirm video, overlays, audio, and stop/start behavior.

16. OBS backup and maintenance plan

Backups are the difference between a quick restore and a full rebuild. The OBS configuration should be backed up whenever scenes, sources, profile settings, or plugin layout changes are made.

Backup item	How / where	When
OBS Profile	OBS Profile menu export or %APPDATA%\obs-studio\basic\profiles	After stream-key/output/audio setting changes.
OBS Scene Collection	OBS Scene Collection menu export or %APPDATA%\obs-studio\basic\scenes	After scene/source/layout changes.
Stream Agent config	Copy config_overrides.json and current app .py file	After Web HUD config changes.
Intro media	Copy intro video file and source folder	After replacing intro video.
Task Scheduler docs	Keep current DOCX/PDF and screenshots	After app path/name changes.
Plugin installers	Save OBS and DistroAV/NDI installers or version notes; save ASIO plugin information only if the fallback ASIO path is used	After stable system verification.

Backup rule

After any successful Sunday with a changed setup, make a backup before experimenting again.

17. Troubleshooting

Symptom	Likely causes	Checks / fixes
Stream Agent says OBS offline	OBS not running; WebSocket disabled; wrong port/password	Open OBS; check Tools/Settings for WebSocket server; verify port 4455 and password.
Director preview blank/stale	Wrong source name; camera offline; OBS screenshot error	Verify West_axis/East_axis exact names; confirm camera feed visible in OBS.
Slides/overlay missing	Proclaim not On Air; NDI source unavailable; source hidden or below camera	Confirm Proclaim On Air; verify slides source in OBS; place above camera layer.
Camera video black	Axis camera offline; RTSP URL wrong; PoE/network issue	Ping camera; open camera web page; check PoE switch; verify source properties.
Audio missing in embedded mode	Camera audio not present; OBS source muted; wrong AUDIO_MODE	Check West_axis/East_axis meters; unmute; confirm Stream Agent Audio / Sync config.
Sync page missing	AUDIO_MODE is axis_embedded	Expected behavior. Switch to asio_shared only if ASIO_audio is in OBS.
Long wait before cut	Preset blind delay applied	Use Director countdown; adjust Preset Delays if delay is too long.
YouTube not live after OBS start	No scheduled placeholder; auto-start off; wrong stream key/profile	Check YouTube Live Control Room and OBS profile NHLC<space>.
Wrong YouTube destination	Wrong OBS profile or stream key	Use profile safety; verify Settings -> Stream before service.
Multiple OBS copies or odd behavior	Task Scheduler or manual launch started duplicate app	Close duplicates; keep Task Scheduler set to not start a new instance.

18. Items to capture later with screenshots

This document is complete enough to guide a rebuild, but it should later be improved with actual OBS screenshots from the working BeelinkOBS setup. Add the following when available:

- OBS main window showing Scenes and Sources for Introduction, West View, and East View.
- Settings -> Stream page with stream key redacted.
- Settings -> Output page.
- Settings -> Audio page showing 48 kHz sample rate and selected devices.
- Settings -> Video page.
- WebSocket server settings page with password redacted.
- West_axis and East_axis source properties with credentials/URLs redacted where necessary.
- slides NDI source properties.
- Audio Mixer Advanced Audio Properties showing sync offsets only if the optional ASIO fallback mode is used.

Appendix A - Configuration record sheet

Configuration item	Expected value / note
OBS version	Record exact working version, currently OBS 32.x family observed.
OBS Profile	NHLC<space>
Scene Collection	Record exact final name after verification.
Streaming server	YouTube RTMP/RTMPS server configured in OBS; do not publish stream key.
Video base/output resolution	Record exact working values after OBS screenshot capture.
FPS	Record exact working value after OBS screenshot capture.
Audio sample rate	48 kHz expected.
WebSocket host/port	127.0.0.1:4455
Scene transition	Fade 400 ms currently acceptable; app may force Cut for selected operations.
West camera source	West_axis
East camera source	East_axis
Proclaim overlay source	slides
Intro media source	Intro_Video
Optional ASIO fallback source	ASIO_audio only if optional AUDIO_MODE = asio_shared

Appendix B - OBS and Stream Agent interface map

Stream Agent function	OBS object touched	Reason
Start Stream	OBS streaming output	Begin sending final program to YouTube.
Stop Stream	OBS streaming output	End encoder output after stop countdown.
Record Toggle	OBS recording output	Optional local recording.
Intro sequence	Introduction scene and	Start stream with intro, wait

	Intro_Video media source	for media end, cut to live scene.
Current view routing	West View / East View scenes	Cut to the scene containing the prepared camera.
Director preview	West_axis / East_axis screenshots	Show camera confidence views in browser.
Audio master fader	West_axis/East_axis embedded audio standard; ASIO_audio only fallback	Control live audio target(s) from Web HUD.
Sync page	ASIO_audio sync offset only if fallback ASIO mode is selected only if fallback ASIO mode is selected	Adjust lip-sync only when optional fallback ASIO mode is active.
Profile safety	OBS active profile	Reduce risk of starting with wrong YouTube profile/key.

Appendix C - Glossary

Term	Meaning in this system
OBS	Open Broadcaster Software, the encoder and scene/source production app.
Scene	An OBS layout containing sources. Stream Agent cuts between scenes.
Source	A video, audio, media, browser, or NDI item inside a scene.
Program	The OBS output currently being streamed/recorded.
WebSocket	Local control channel used by Stream Agent to command OBS.
NDI	Network Device Interface, used here to bring Proclaim overlay video into OBS.
RTSP	Real-Time Streaming Protocol, used by Axis cameras to send video/audio to OBS.
RTMP/RTMPS	Streaming protocol used by OBS to send the final feed to YouTube.
ASIO	Audio Stream Input/Output, retained only as an optional fallback/legacy audio mode.
Blind delay	A wait before cutting to a moved camera so viewers do not see the PTZ motion.

Appendix D - References

These sources were used for general OBS/YouTube/plugin behavior. Local New Hope values come from the Stream Agent III sd app configuration and prior project records.

- OBS Knowledge Base - Sources Guide: <https://obsproject.com/kb/sources-guide>
- OBS Knowledge Base - Profiles: <https://obsproject.com/kb/profiles>
- OBS Knowledge Base - Scene Collections: <https://obsproject.com/kb/scene-collections>
- OBS Knowledge Base - Remote Control Guide: <https://obsproject.com/kb/remote-control-guide>
- OBS Knowledge Base - Launch Parameters: <https://obsproject.com/kb/launch-parameters>
- DistroAV / OBS NDI plugin project: <https://github.com/DistroAV/DistroAV>
- OBS ASIO plugin project: <https://github.com/Andersama/obs-asio>
- YouTube Help - Manage live stream settings: <https://support.google.com/youtube/answer/9854503>
- YouTube Help - Encoder settings, bitrates, and resolutions:
<https://support.google.com/youtube/answer/2853702>